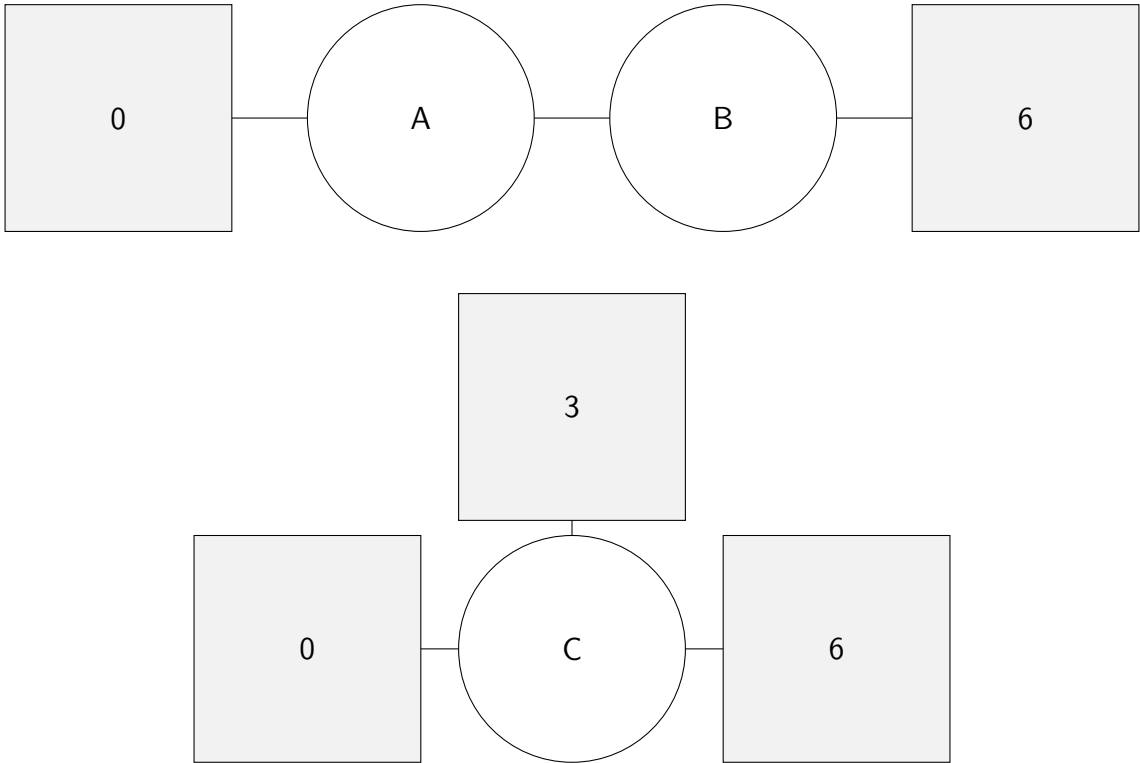


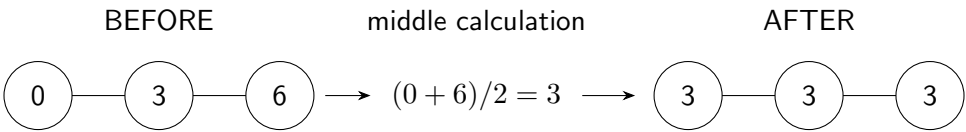
A wandering token starts at a circle. At each move, choose fairly among all joined neighbors. Stop when it first reaches a square and record that square’s score. A square with score 0 is still a possible choice.

Problem 1: Run at least twelve walks starting at A, then twelve starting at B. Use the walks to estimate the average score after many walks from each start. Use the same rule at C on the second board. Find exact expected scores for all three starts.

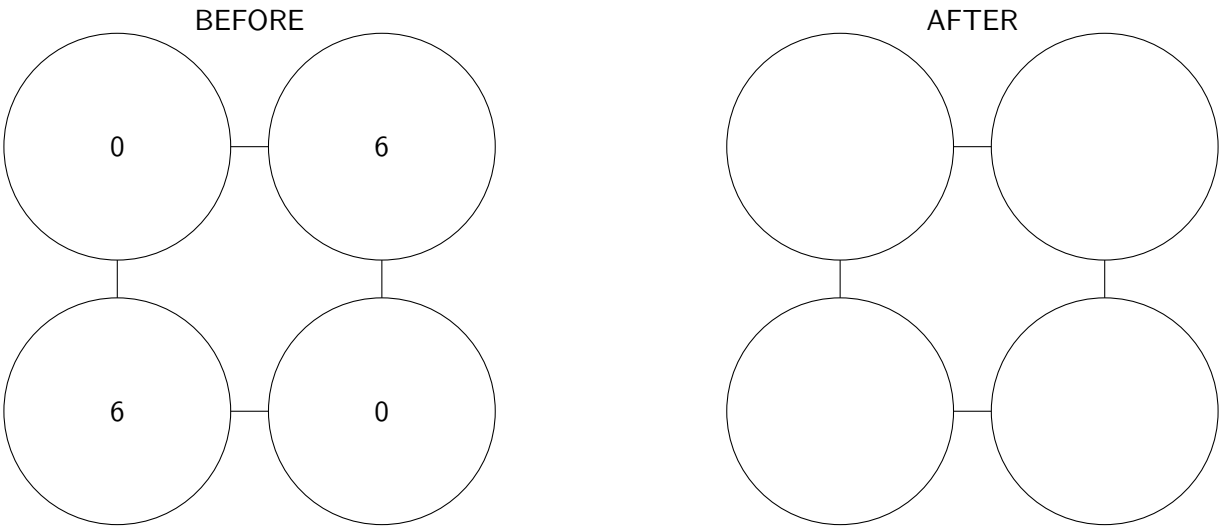


start	scores at the end of the walks
A	
B	
C	

At a tick, every circle takes the average of its neighbors' BEFORE values. Put all new values on a separate AFTER board. Then that whole board becomes the next BEFORE board. No value is fixed.



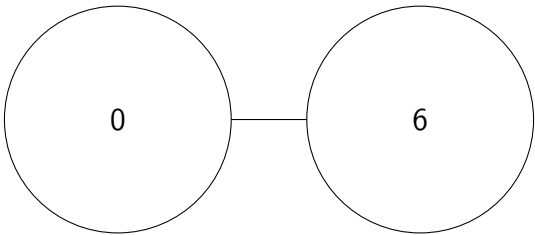
Problem 2: Start with these four values. Keep updating the whole board until a state returns. Find a different starting state that first returns after two ticks.



tick	top left	top right	bottom right	bottom left
0	0	6	0	6
1				
2				
3				
4				

Change the tick rule: average each circle’s own BEFORE value together with all its neighbors’ BEFORE values. Put all replacements on the AFTER board together.

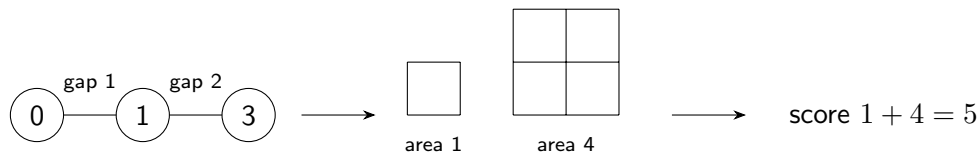
Problem 3: Start with two joined circles, 0 and 6. Compare repeated ticks using the rule from Problem 2 and the new rule above. Does either process settle to one unchanged board?



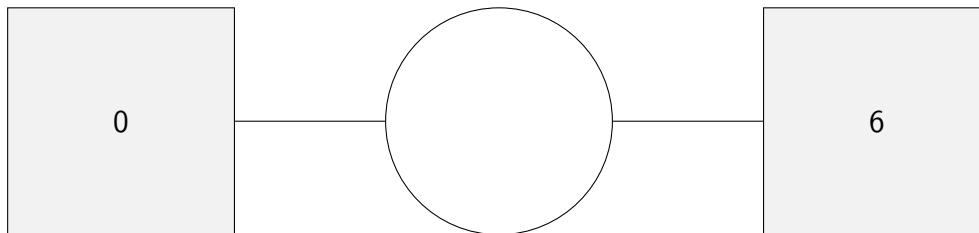
Problem 4: Use the new rule on the four-circle board starting at 0, 6, 0, 6. Find the next three whole boards. What value do you think each circle approaches? Can it reach that value exactly after a finite number of ticks?

tick	top left	top right	bottom right	bottom left
0	0	6	0	6
1				
2				
3				

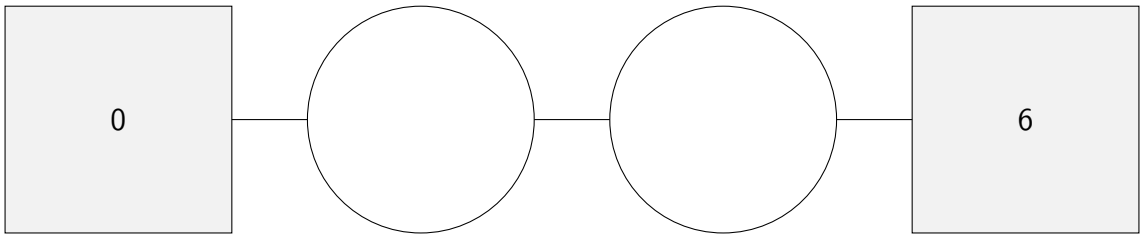
For a roughness score, find the gap between the two values on each edge. Make a square whose side is that gap. Add all the square areas, counting each edge once. Squares on the graph keep their values.



Problem 5: Put a whole number from 0 to 6 at the circle. Find the smallest possible roughness score. Explain why no other choice gives a smaller score.



Problem 6: Choose whole-number values from 0 to 6 for the two circles. Make the roughness score as small as possible. Explain why your best score cannot be beaten.



left circle	right circle	roughness score

